Daniel Engelberg

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Herzliya, Israel | Authorized to work in the US | Relocating to New York

SUMMARY

Mechanical engineer (B.Sc., Technion, robotics) who designs, builds and tests: a MuJoCo simulation in which a robot arm catches a tumbling object in zero gravity (85% of 300 randomized throws), SolidWorks design with strength and thermal analysis of a Formula Student accumulator, and hands-on prototyping and fabrication.

SKILLS

CAD and analysis: SolidWorks, Onshape, Creo, AutoCAD, Kenesto; strength and thermal analysis; BOMs
Hands-on: Welding, grinding, drilling, metal cutting; assembly, testing, troubleshooting, repair
Robotics: MuJoCo, Kalman filtering, inverse kinematics, path planning, control theory, ROS
Programming: Python, JavaScript, TypeScript, Node.js, SQL, MATLAB
Languages: English (native), Hebrew (fluent)

EDUCATION

Technion – Israel Institute of Technology, Haifa, Israel Mar 2026
B.Sc. in Mechanical Engineering, Robotics specialization
Relevant coursework: Mechanical Design, CAD, Manufacturing Processes, Solid Mechanics, Fluid Mechanics, Control Theory, Robotics (Kinematics, Path Planning), Signal Processing, Python

PROJECTS

Zero-G Catch (robotic capture in simulation) | *Python, MuJoCo, NumPy* Oct 2026

- Built a system in MuJoCo for a 7-DoF Franka Panda arm to catch a drifting, tumbling 0.5 kg bar in zero gravity and bring it to rest, sensing it only through a noisy 30 Hz pose sensor (5 mm and 2° noise, 5% dropped readings)
- Caught and stopped 85% of 300 randomized throws (2–15 cm/s drift, 0–30°/s spin) at a median 1.9 cm/s contact speed and 2.5 N peak grip; 99% of grasps that started ended at rest
- Wrote a Kalman filter and spin estimator, an intercept planner with inverse kinematics and singularity margins, and a controller that rides along with the spinning bar, closes only when the estimated miss is under 1 cm and 2 cm/s, and brakes at under 0.3 m/s²
- Traced failures with Monte Carlo testing and fixed a planner that committed to a single grasp and never retried, raising the catch rate from 65% to 85%; published at zero-g-catch.vercel.app

Automatic Cheese Grater (mechanical design)

- Designed the mechanism in CAD with 3D assemblies, engineering drawings and a bill of materials
- Resolved manufacturability, tolerance, assembly, cost and safety trade-offs in the design

Rolling Chassis Fabrication

- Fabricated a rolling chassis from steel pipe and scrap: cut stock with a belt cutter and angle grinder, drilled on an industrial drill, welded and ground the joints, and assembled the frame
- Diagnosed and repaired machinery and fabricated metal parts in a family engineering workshop; restored a vintage car

Hotel Robot Operating System

- Built a hotel operations system that dispatches delivery and logistics robots, coordinates elevator rides and takes requests from the property management system (PMS)

EXPERIENCE

FlowTransit, Tel Aviv, Israel Jul 2026 – Sep 2026
Intern

- Studied 4 route-planning approaches (reinforcement learning with PPO, genetic algorithms, simulated annealing, shortest-path methods) to compare how each evaluates transit networks
- Improved GTFS and OpenStreetMap preprocessing so complete intracity bus lines are kept whenever any segment crosses a selected area, using map matching and network edge processing

Formula Student Technion, Haifa, Israel Oct 2023 – Apr 2024
High Voltage Engineer

- Co-designed a new high-voltage accumulator for the race car from scratch in SolidWorks, managing files in Kenesto and tasks in monday.com
- Ran strength analysis and designed the accumulator's wiring architecture and thermal management in SolidWorks
- Compiled specifications for every component and built the accumulator's bill of materials (BOM) in Excel